# Network Monitoring and Wireshark

CSE 322

# Acknowledgement

- https://studylib.net/doc/17803129/cap6135--malware-and-software-vulnerability-analysis-netw...
- https://slideplayer.com/slide/5692294/
- http://ilta.ebiz.uapps.net/ProductFiles/productfiles/672/
  wireshark.ppt

- UC Berkley course “EE 122: Intro to Communication Networks”
  - http://www.eecs.berkeley.edu/~jortiz/courses/ee122/presentations/Wireshark.ppt
- Other resources:
  - http://openmaniak.com/wireshark_filters.php

# Motivation for Network Monitoring

- Essential for Network Management
  - Router and Firewall policy
  - Detecting abnormal/error in networking
  - Access control
- Security Management
  - Detecting abnormal traffic
  - Traffic log for future forensic analysis

# Tools Overview

- Tcpdump
  - Unix-based command-line tool used to intercept packets
    - Including filtering just the packets of interest
  - Reads “live traffic” from interface specified using **`-i`** option …
  - … or from a previously recorded *trace file* specified using **`-r`** option
    - You create these when capturing live traffic using **`-w`** option
- Tshark
  - Tcpdump-like capture program
  - Very similar behavior and flags to tcpdump
- Wireshark
  - GUI for displaying tcpdump/tshark packet traces

# tcpdump Example

- Ran tcpdump on a Unix machine
- Example: first few lines of the output

```
01:46:28.808262 IP danjo.CS.Berkeley.EDU.ssh > adsl-69-228-230-
    7.dsl.pltn13.pacbell.net.2481: . 2513546054:2513547434(1380)
    ack 1268355216 win 12816
01:46:28.808271 IP danjo.CS.Berkeley.EDU.ssh > adsl-69-228-230-
    7.dsl.pltn13.pacbell.net.2481: P 1380:2128(748) ack 1 win 12816
01:46:28.808276 IP danjo.CS.Berkeley.EDU.ssh > adsl-69-228-230-
    7.dsl.pltn13.pacbell.net.2481: . 2128:3508(1380) ack 1 win 12816
01:46:28.890021 IP adsl-69-228-230-7.dsl.pltn13.pacbell.net.2481 >
    danjo.CS.Berkeley.EDU.ssh: P 1:49(48) ack 1380 win 16560
```

# What Does a Line Convey?

Timestamp

This is an IP packet

Source host name

Source port number (22)

01:46:28.808262 IP danjo.CS.Berkeley.EDU.ssh > adsl-69-228-230-7.dsl.pltn13.pacbell.net.2481: . 2513546054:2513547434(1380) ack 1268355216 win 12816

Destination host name

Destination port number

TCP specific information

- Different output formats for different packet types

# Output from tcpdump

```
manav@ubuntulinux:~$ sudo tcpdump
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on wlo1, link-type EN10MB (Ethernet), capture size 262144 bytes
23:12:15.734637 IP6 fe80::6c8:7ff:fe26:ceac > ff02::16: HBH ICMP6, multicast listener report v2, 1 group record(s), length 28
23:12:15.737024 IP ubuntulinux.60811 > b.resolvers.Level3.net.domain: 60873+ PTR? 6.1.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.2.0.f.f.ip6.arpa. (90)
23:12:15.937343 IP b.resolvers.Level3.net.domain > ubuntulinux.60811: 60873 NXDomain 0/1/0 (154)
23:12:15.939727 IP ubuntulinux.44132 > b.resolvers.Level3.net.domain: 7027+ PTR? c.a.e.c.6.2.e.f.f.f.7.0.8.c.6.0.0.0.0.0.0.0.0.0.0.0.0.0.0.8.e.f.ip6.arpa. (90)
23:12:16.142628 IP b.resolvers.Level3.net.domain > ubuntulinux.44132: 7027 NXDomain* 0/1/0 (149)
23:12:16.144477 IP ubuntulinux.54078 > b.resolvers.Level3.net.domain: 44074+ PTR? 2.2.2.4.in-addr.arpa. (38)
23:12:16.182985 IP6 fe80::6c8:7ff:fe26:ceac.mdns > ff02::fb.mdns: 0 [2q] [2n] ANY (QU)? Android.local. ANY (QU)? Android.local. (81)
23:12:16.346940 IP b.resolvers.Level3.net.domain > ubuntulinux.54078: 44074 1/0/0 PTR b.resolvers.Level3.net. (74)
23:12:16.348070 IP ubuntulinux.40932 > b.resolvers.Level3.net.domain: 34612+ PTR? 102.0.168.192.in-addr.arpa. (44)
23:12:16.442544 IP6 fe80::6c8:7ff:fe26:ceac.mdns > ff02::fb.mdns: 0 [2q] [2n] ANY (QM)? Android.local. ANY (QM)? Android.local. (81)
23:12:16.515363 IP b.resolvers.Level3.net.domain > ubuntulinux.40932: 34612 NXDomain* 0/1/0 (103)
23:12:16.517489 IP ubuntulinux.53519 > b.resolvers.Level3.net.domain: 3094+ PTR? b.f.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.2.0.f.f.ip6.arpa. (90)
23:12:16.590778 IP6 fe80::6c8:7ff:fe26:ceac > ff02::16: HBH ICMP6, multicast listener report v2, 1 group record(s), length 28
23:12:16.680420 IP b.resolvers.Level3.net.domain > ubuntulinux.53519: 3094 NXDomain 0/1/0 (154)
23:12:16.683273 IP6 fe80::6c8:7ff:fe26:ceac.mdns > ff02::fb.mdns: 0 [2q] [2n] ANY (QM)? Android.local. ANY (QM)? Android.local. (81)
23:12:16.961783 IP6 fe80::6c8:7ff:fe26:ceac.mdns > ff02::fb.mdns: 0*- [0q] 4/0/3 (Cache flush) PTR Android.local., (Cache flush) PTR Android.local., (Cache flush) A 192.168.0.101, (Cache flush) AAAA fe80::6c8:7f
f:fe26:ceac (247)
23:12:17.986627 IP6 fe80::6c8:7ff:fe26:ceac.mdns > ff02::fb.mdns: 0*- [0q] 4/0/3 (Cache flush) PTR Android.local., (Cache flush) PTR Android.local., (Cache flush) A 192.168.0.101, (Cache flush) AAAA fe80::6c8:7f
f:fe26:ceac (247)
23:12:18.238252 IP ubuntulinux.35076 > a23-39-122-85.deploy.static.akamaitechnologies.com.https: Flags [.], ack 89677080, win 501, options [nop,nop,TS val 2010432084 ecr 2402112534], length 0
23:12:18.239048 IP ubuntulinux.55784 > b.resolvers.Level3.net.domain: 49805+ PTR? 85.122.39.23.in-addr.arpa. (43)
23:12:18.497830 IP a23-39-122-85.deploy.static.akamaitechnologies.com.https > ubuntulinux.35076: Flags [R], seq 89677080, win 0, length 0
23:12:18.497830 IP b.resolvers.Level3.net.domain > ubuntulinux.55784: 49805 1/0/0 PTR a23-39-122-85.deploy.static.akamaitechnologies.com. (107)
23:12:19.101841 IP ubuntulinux.55857 > 239.255.255.250.1900: UDP, length 171
23:12:19.102591 IP ubuntulinux.49879 > b.resolvers.Level3.net.domain: 54682+ PTR? 250.255.255.239.in-addr.arpa. (46)
23:12:19.317004 IP b.resolvers.Level3.net.domain > ubuntulinux.49879: 54682 NXDomain 0/1/0 (103)
23:12:19.636296 IP ubuntulinux.bootpc > _gateway.bootps: BOOTP/DHCP, Request from 84:fd:d1:e5:20:5e (oui Unknown), length 289
23:12:19.637071 IP ubuntulinux.51783 > b.resolvers.Level3.net.domain: 29099+ PTR? 1.0.168.192.in-addr.arpa. (42)
23:12:19.831442 IP _gateway.bootps > 255.255.255.255.bootpc: BOOTP/DHCP, Reply, length 295
23:12:19.931402 IP b.resolvers.Level3.net.domain > ubuntulinux.51783: 29099 NXDomain* 0/1/0 (101)
23:12:20.102446 IP ubuntulinux.55857 > 239.255.255.250.1900: UDP, length 171
^C23:12:20.239751 IP 192.168.0.3.mdns > 224.0.0.251.mdns: 16 [2q] PTR (QM)? _233637DE._sub._googlecast._tcp.local. PTR (QM)? _googlecast._tcp.local. (61)

30 packets captured
34 packets received by filter
0 packets dropped by kernel
manav@ubuntulinux:~$
```

- Link: https://www.geeksforgeeks.org/tcpdump-command-in-linux-with-examples/

# Similar Output from Tshark

```
1190003744.940437 61.184.241.230 -> 128.32.48.169
    SSH Encrypted request packet len=48
1190003744.940916 128.32.48.169 -> 61.184.241.230
    SSH Encrypted response packet len=48
1190003744.955764 61.184.241.230 -> 128.32.48.169
    TCP 6943 > ssh [ACK] Seq=48 Ack=48 Win=65514
    Len=0 TSV=445871583 TSER=632535493
1190003745.035678 61.184.241.230 -> 128.32.48.169
    SSH Encrypted request packet len=48
1190003745.036004 128.32.48.169 -> 61.184.241.230
    SSH Encrypted response packet len=48
1190003745.050970 61.184.241.230 -> 128.32.48.169
    TCP 6943 > ssh [ACK] Seq=96 Ack=96 Win=65514
    Len=0 TSV=445871583 TSER=632535502
```

# Demo 1 – Basic Run

- Syntax:

  *tcpdump [options] [filter expression]*

- To do -
  - $ sudo tcpdump –i eth0
    - Sudo command allows you to run tcpdump in root privilege
  - On your own Unix machine, you can run it using “sudo” or directly run “tcpdump” if you have root privilege
- Observe the output

# Filters

- We are often not interested in all packets flowing through the network
- Use filters to capture only packets of interest to us

# Demo 2

1. Capture only udp packets
   - tcpdump “udp”
2. Capture only tcp packets
   - tcpdump “tcp”

# Demo 2 (contd.)

1. Capture only UDP packets with destination port 53 (DNS requests)
   - tcpdump “udp dst port 53”
2. Capture only UDP packets with source port 53 (DNS replies)
   - tcpdump “udp src port 53”
3. Capture only UDP packets with source or destination port 53 (DNS requests and replies)
   - tcpdump “udp port 53”

# Running tcpdump

- Requires superuser/administrator privileges on Unix
  - http://www.tcpdump.org/
  - You can do it on your own Unix machine
  - You can install a Linux OS in Vmware on your windows machine

- Tcpdump for Windows
  - WinDump: http://www.winpcap.org/windump/
    - Free software
    - Might have compatibility issues

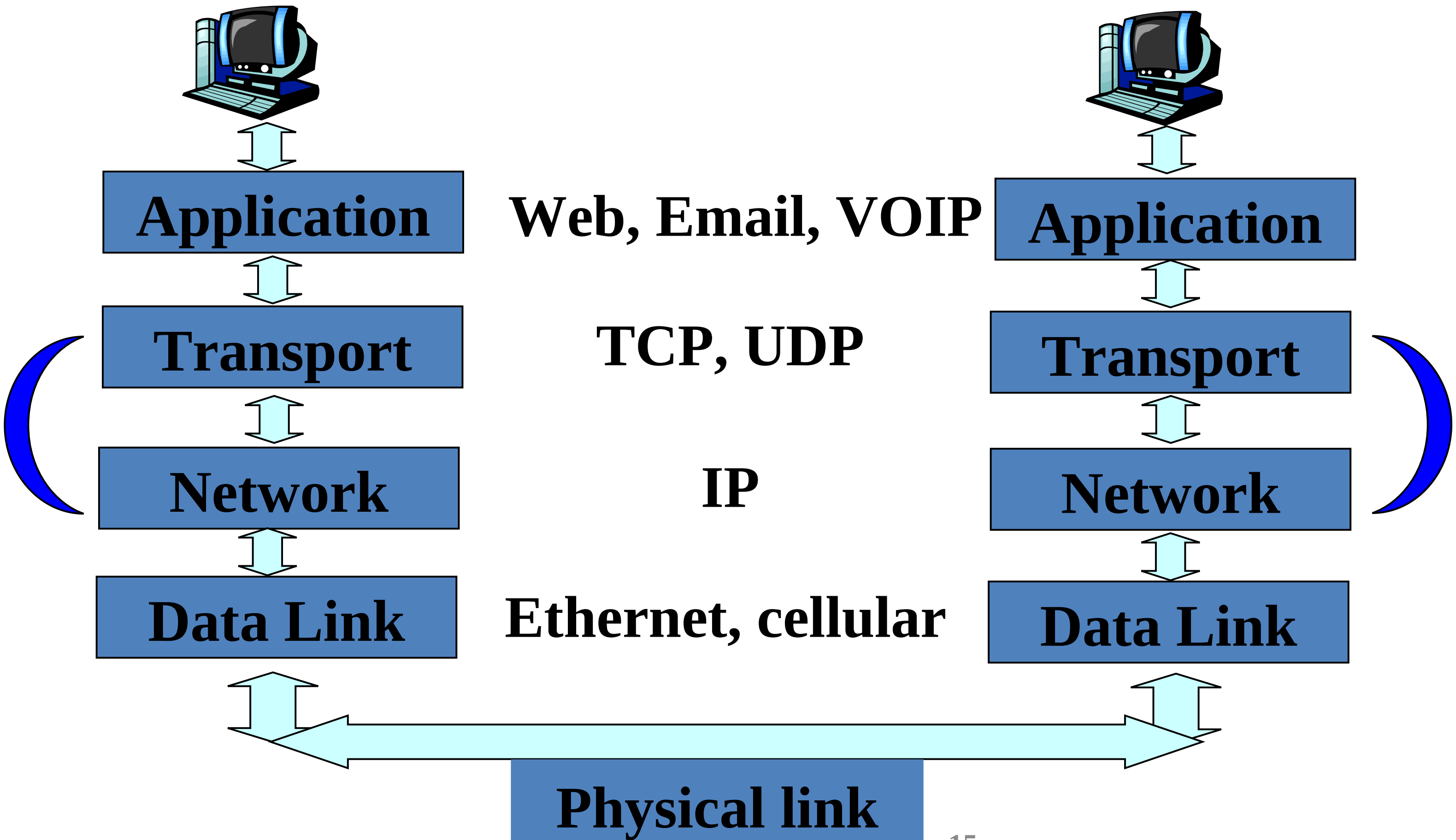
# So What is WireShark?

- Packet sniffer/protocol analyzer
- Open Source Network Tool
- Latest version of the ethereal tool

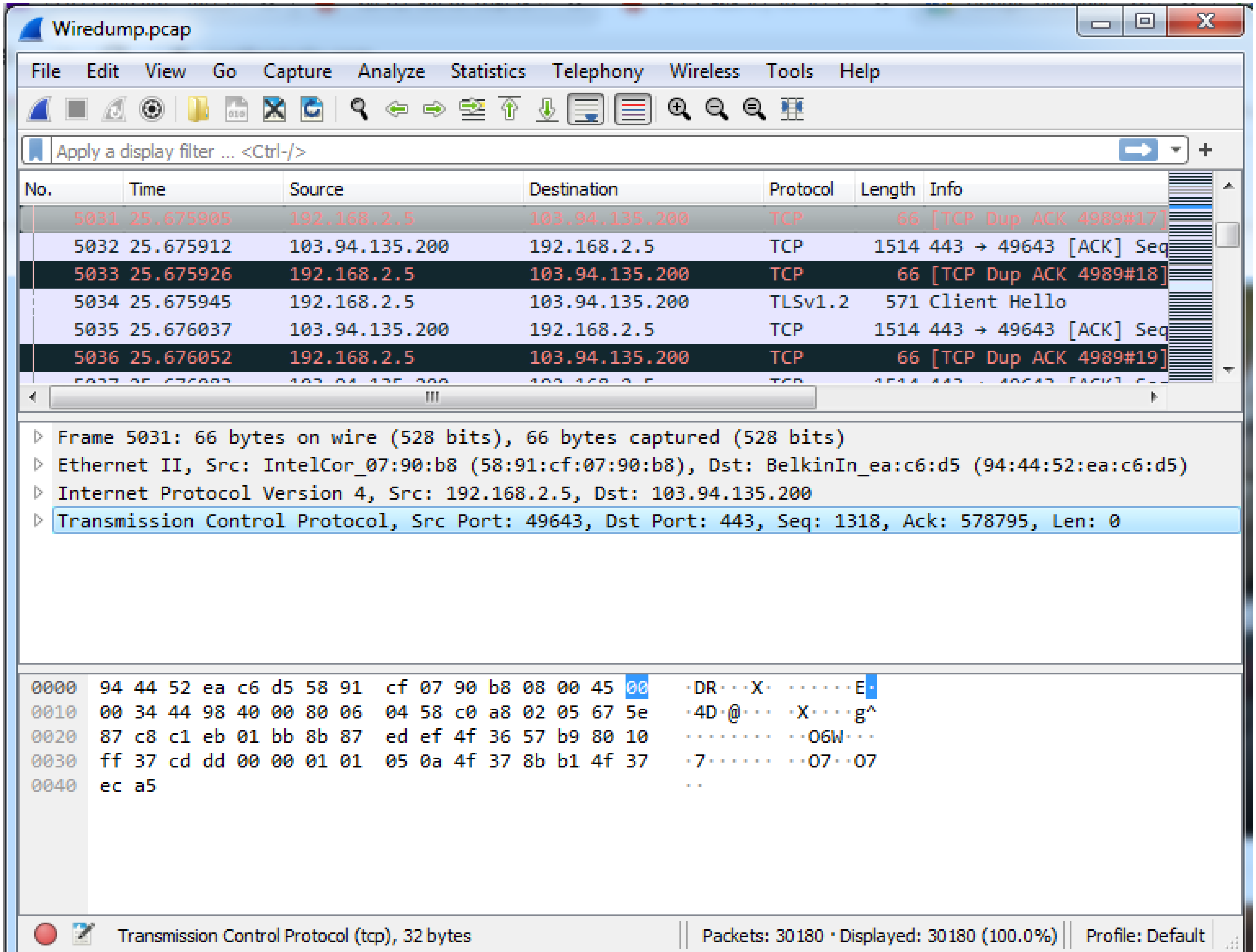
| # | Wireshark | Tcpdump |
|---|---|---|
| **1** | A GUI tool to catch data packets | A CLI-based packet capturing tool |
| **2** | It does packet analysis, decode data payloads if the encryption keys are identified, and recognize data payloads from file transfers such as smtp, http, etc. | Tcpdump only provides do a simple analysis of such types of traffic, such as DNS queries |
| **3** | It has advanced network interfaces | It has conventional interfaces |
| **4** | Good for complex filters | Used for simple filters |
| **5** | It provides decoding of protocol- | It is less efficient in decoding |

# Scope of WireShark

- Internet and all …



# Wireshark Interface



# Wireshark Interface – Different Parts

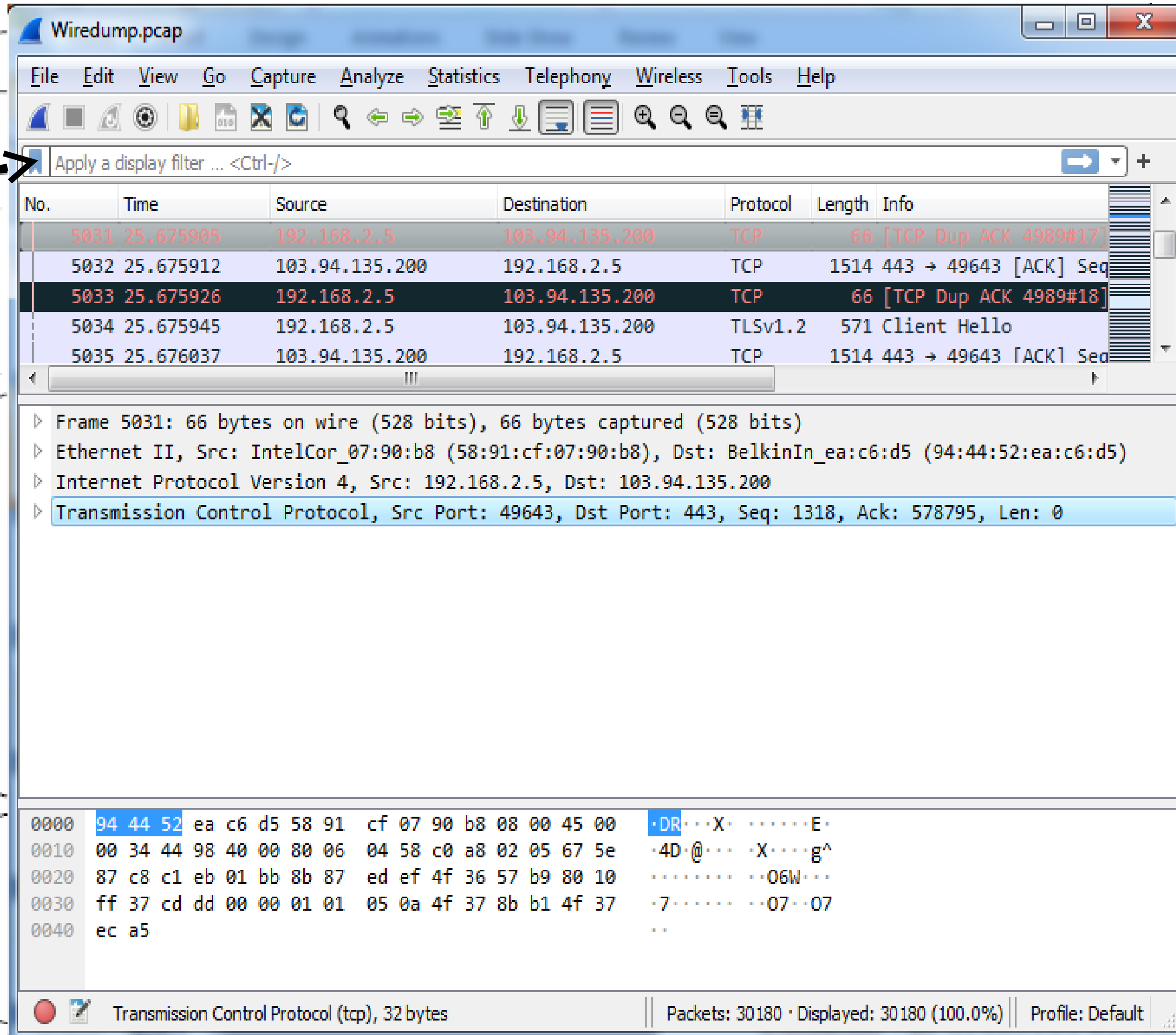
command menus

display filter specification

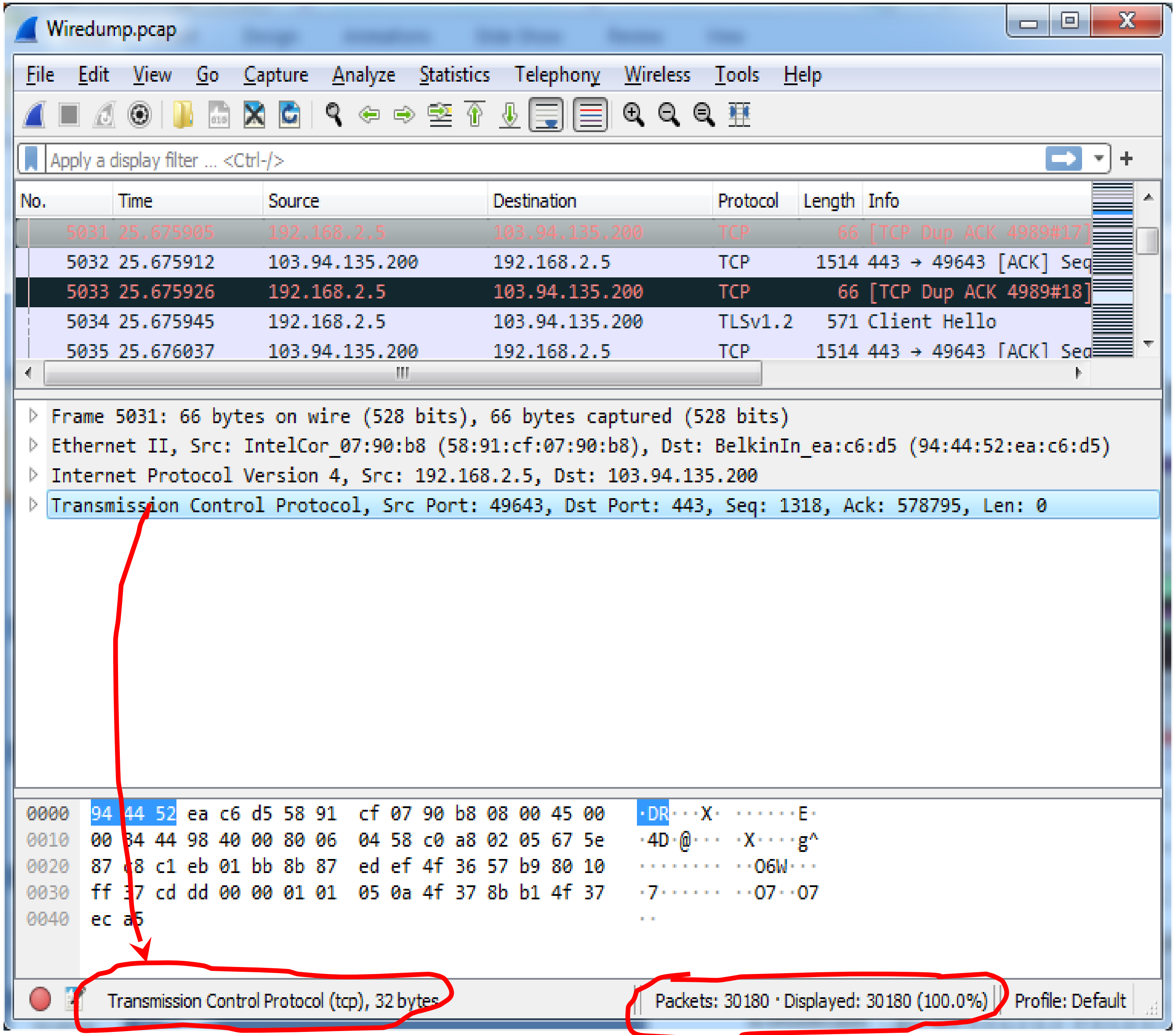
listing of captured packets

details of selected packet header

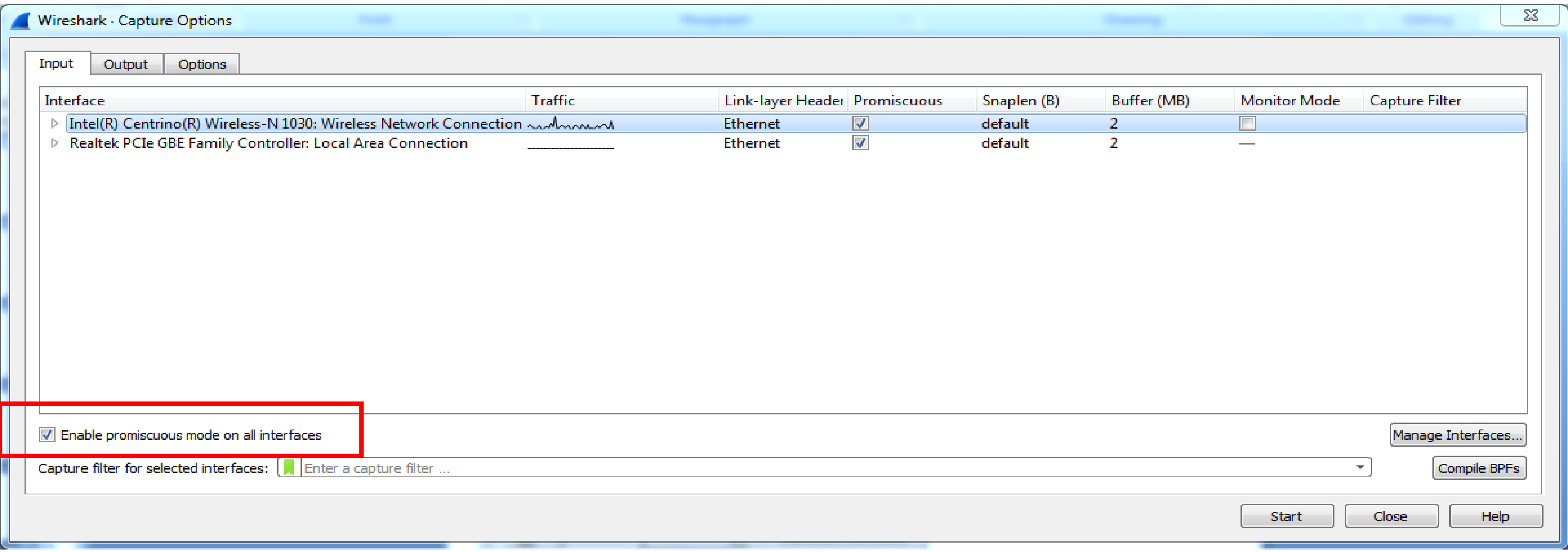
packet content in hexadecimal and ASCII



# Status Bar


Wiredump.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
5031 25.675905 192.168.2.5 103.94.135.200 TCP 66 [TCP Dup ACK 4989#17]
5032 25.675912 103.94.135.200 192.168.2.5 TCP 1514 443 → 49643 [ACK] Seq
5033 25.675926 192.168.2.5 103.94.135.200 TCP 66 [TCP Dup ACK 4989#18]
5034 25.675945 192.168.2.5 103.94.135.200 TLSv1.2 571 Client Hello
5035 25.676037 103.94.135.200 192.168.2.5 TCP 1514 443 → 49643 [ACK] Seq
Frame 5031: 66 bytes on wire (528 bits), 66 bytes captured (528 bits)
Ethernet II, Src: IntelCor_07:90:b8 (58:91:cf:07:90:b8), Dst: BelkinIn_ea:c6:d5 (94:44:52:ea:c6:d5)
Internet Protocol Version 4, Src: 192.168.2.5, Dst: 103.94.135.200
Transmission Control Protocol, Src Port: 49643, Dst Port: 443, Seq: 1318, Ack: 578795, Len: 0
0000 94 44 52 ea c6 d5 58 91 cf 07 90 b8 08 00 45 00
0010 00 34 44 98 40 00 80 06 04 58 c0 a8 02 05 67 5e
0020 87 c8 c1 eb 01 bb 8b 87 ed ef 4f 36 57 b9 80 10
0030 ff 37 cd dd 00 00 01 01 05 0a 4f 37 8b b1 4f 37
0040 ec a5
Transmission Control Protocol (tcp), 32 bytes
Packets: 30180 · Displayed: 30180 (100.0%)
Profile: Default

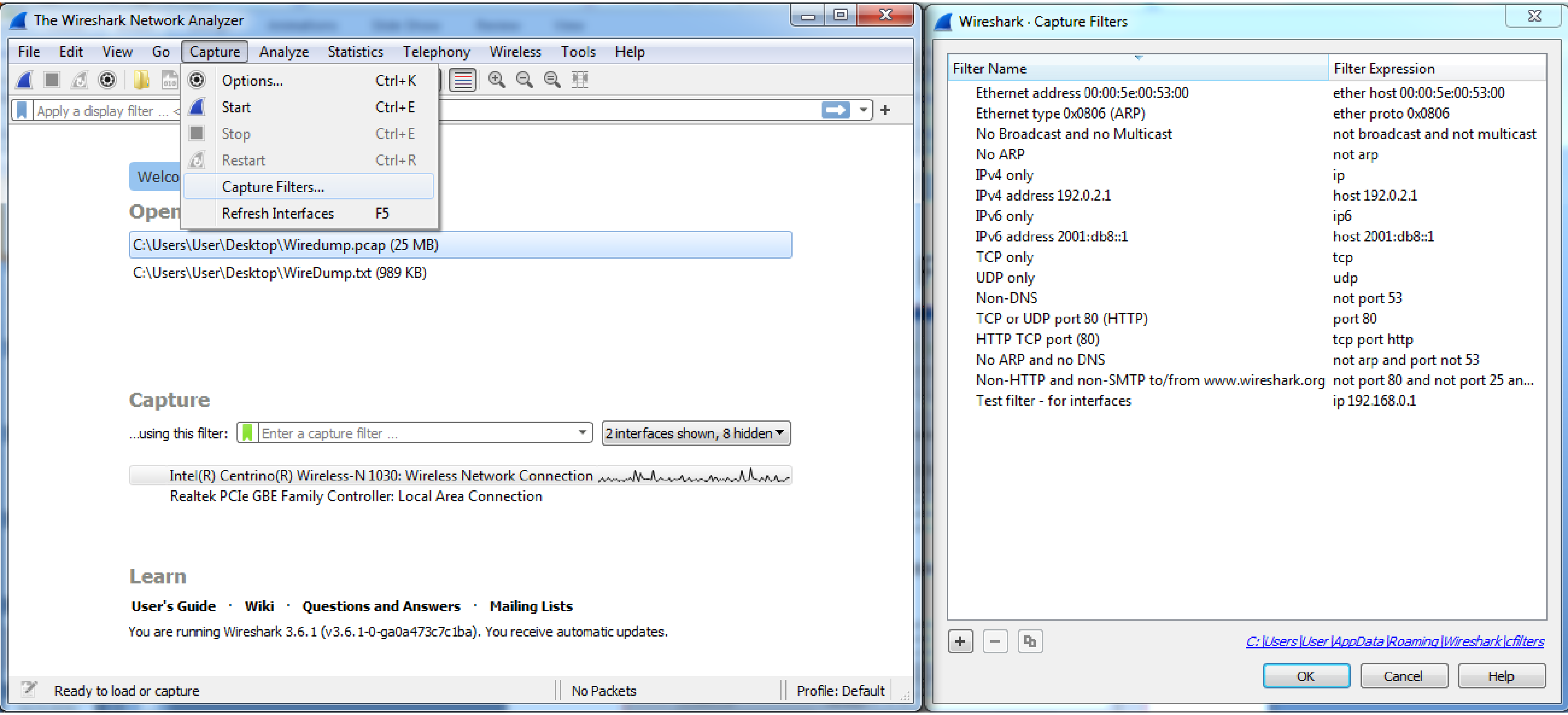
# Capture Options




- Promiscuous mode is used to capture all traffic

- Sometime this does not work
  - Driver does not support
  - You are on a switched LAN

# Capture Filter



# Capture Filter Examples

host 10.1.11.24

host 192.168.0.1 and host 10.1.11.1

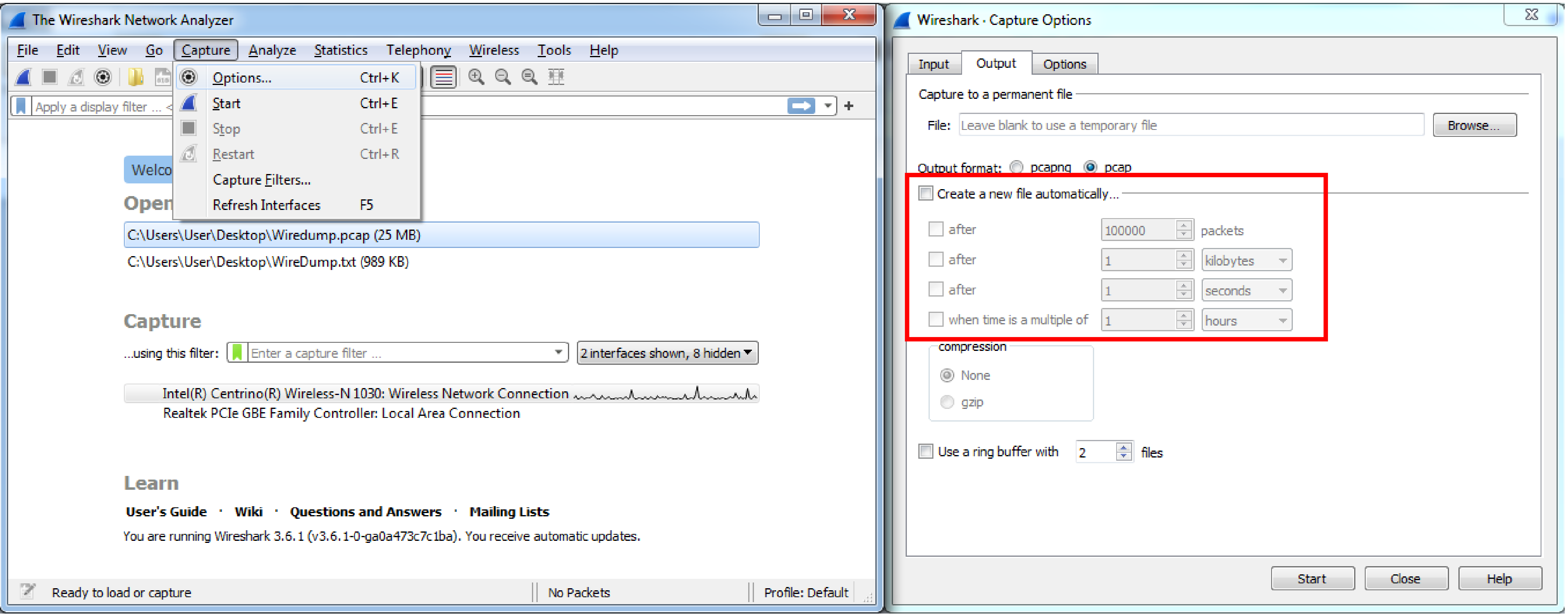
tcp port http

ip

not broadcast not multicast

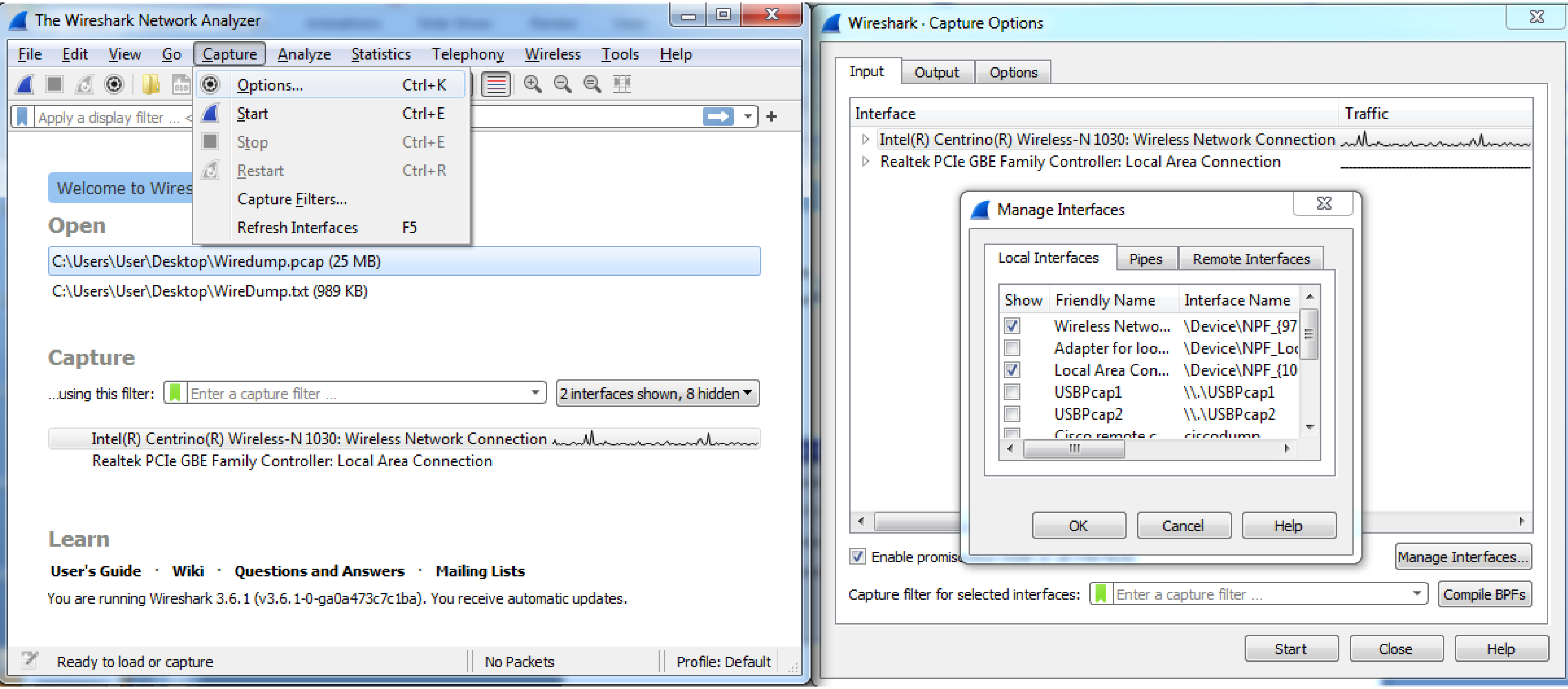
ether host 00:04:13:00:09:a3

# Capture Options

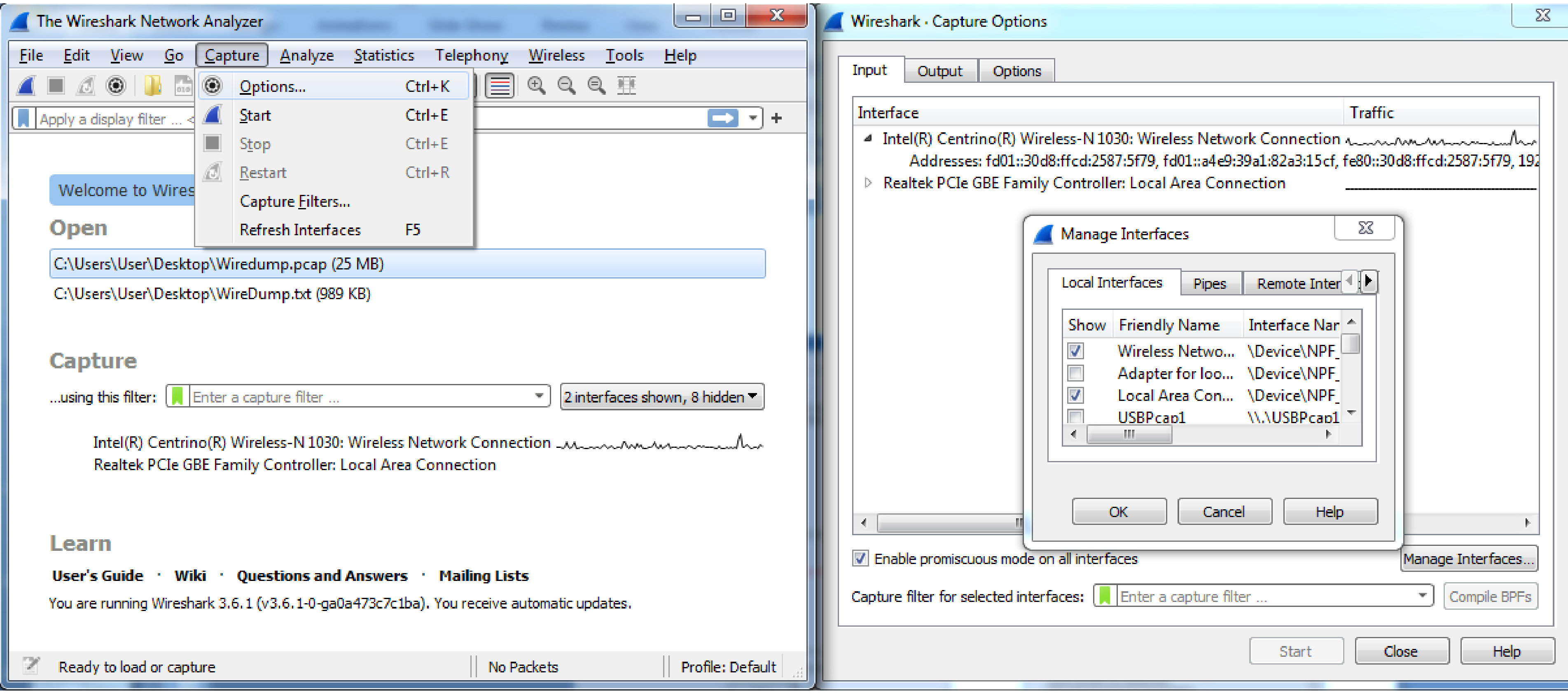



- Ring buffer
  - Addresses a common issue many analysts encounter when capturing packets - huge traces
  - When you use a Ring Buffer, you can define how many files you want to capture and various parameters that affects the file size (# of packets, bytes, and time)
- Link: https://www.networkcomputing.com/networking/working-ring-buffer-wireshark-files

# Capture Interfaces



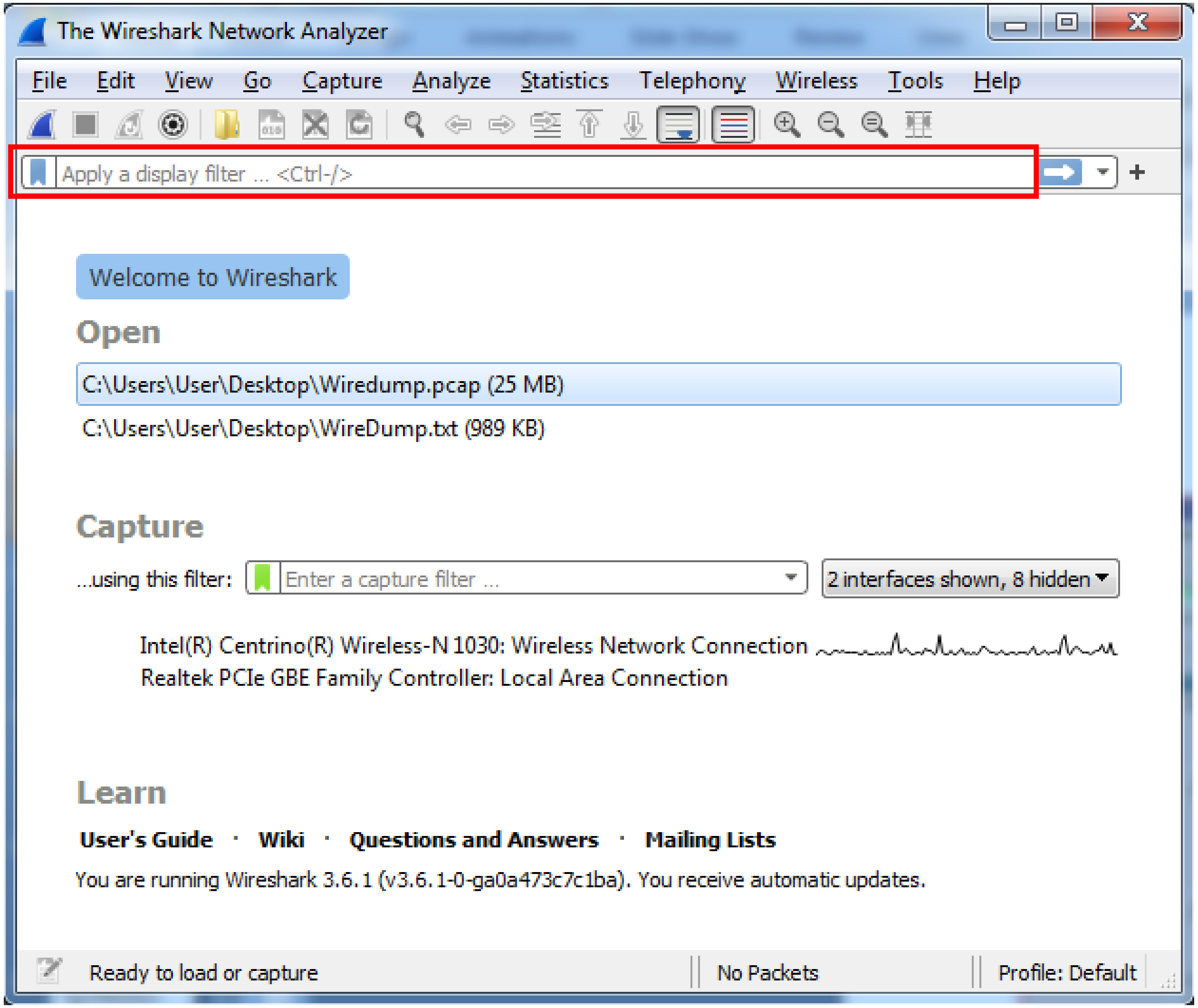
# Interface Details



# Display Filters (Post-Filters)

- Display filters (also called post-filters) only filter the view of what you are seeing
  - In analysis
  - All packets in the capture still exist in the trace
- Display filters use their own format and are much more powerful then capture filters

# Display Filter



# Display Filter Examples

ip.src==10.1.11.00/24

ip.addr==192.168.1.10 && ip.addr==192.168.1.20

tcp.port==80 || tcp.port==3389

!(ip.addr==192.168.1.10 && ip.addr==192.168.1.20)

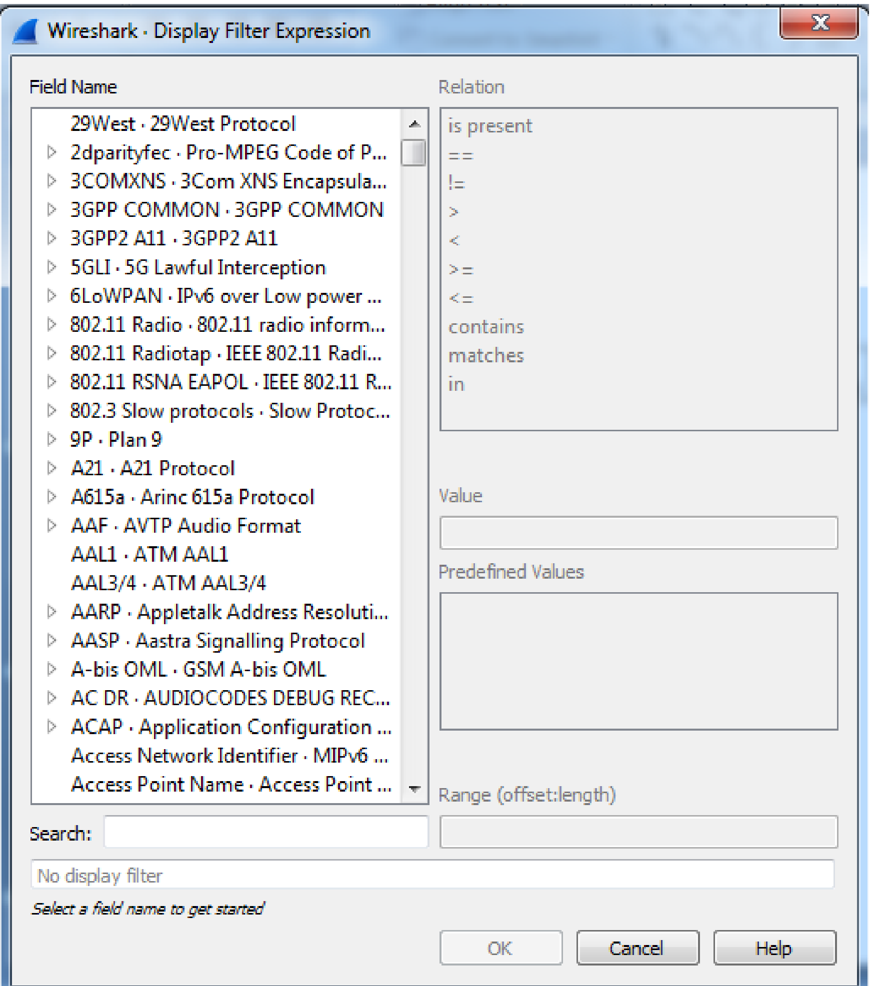
(ip.addr==192.168.1.10 && ip.addr==192.168.1.20) &&
(tcp.port==445 || tcp.port==139)

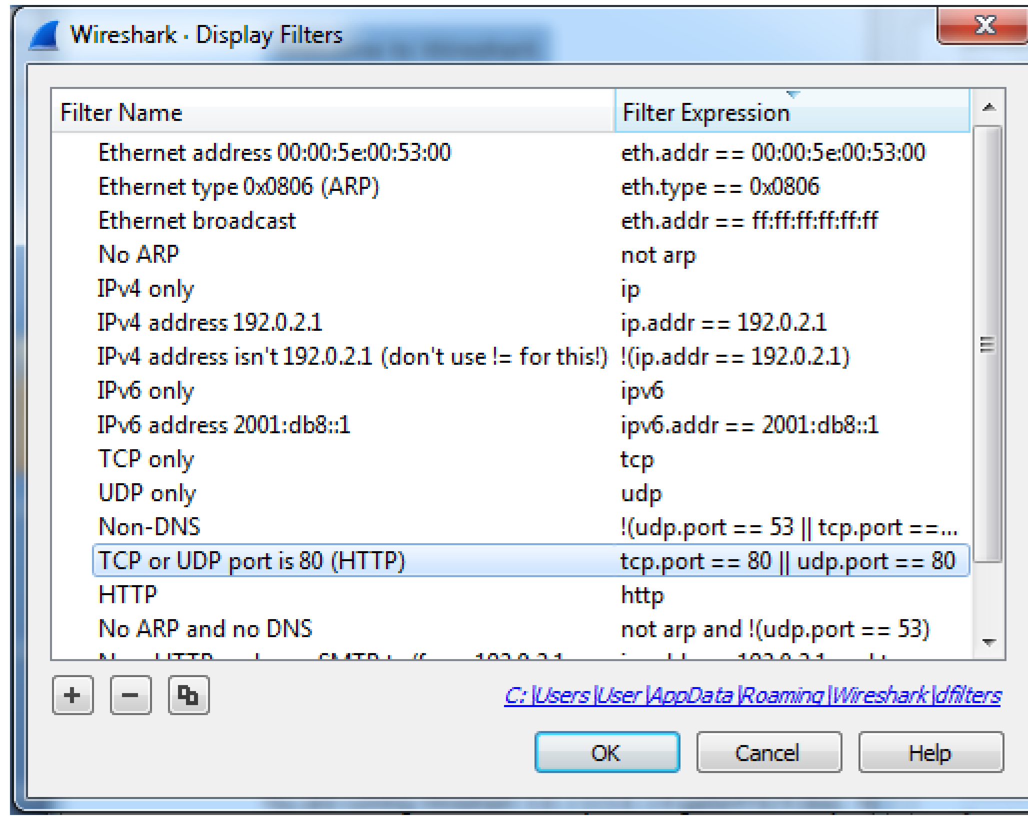
(ip.addr==192.168.1.10 && ip.addr==192.168.1.20) && (udp.port==67
|| udp.port==68)

tcp.dstport == 80

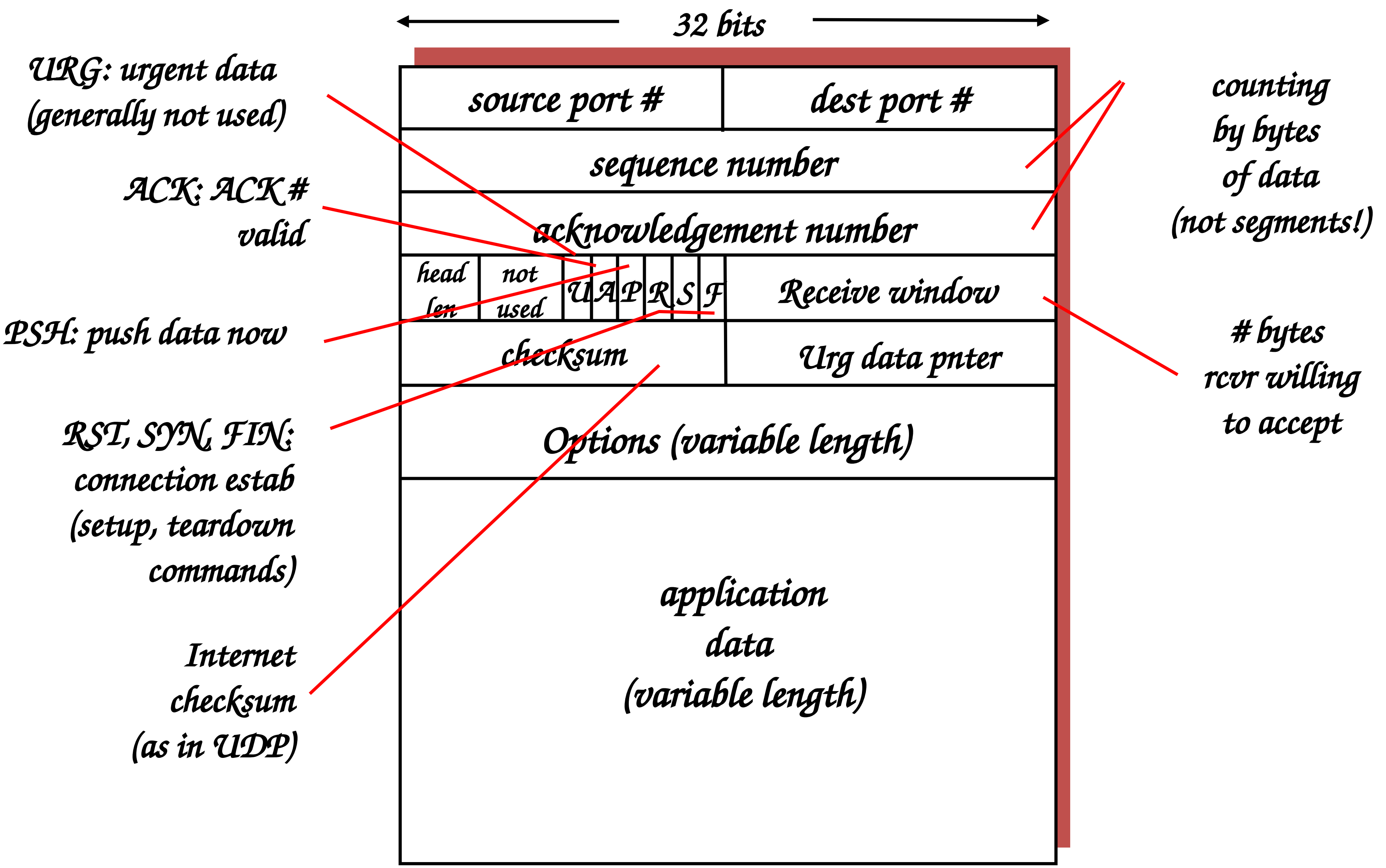
# Display Filter

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Syntax: | **Protocol** . | **String 1** . | **String 2** | **Comparison operator** | **Value** | **Logical Operations** | **Other expression** |
| Example: | ftp | passive | ip | == | 10.2.3.4 | xor | icmp.type |



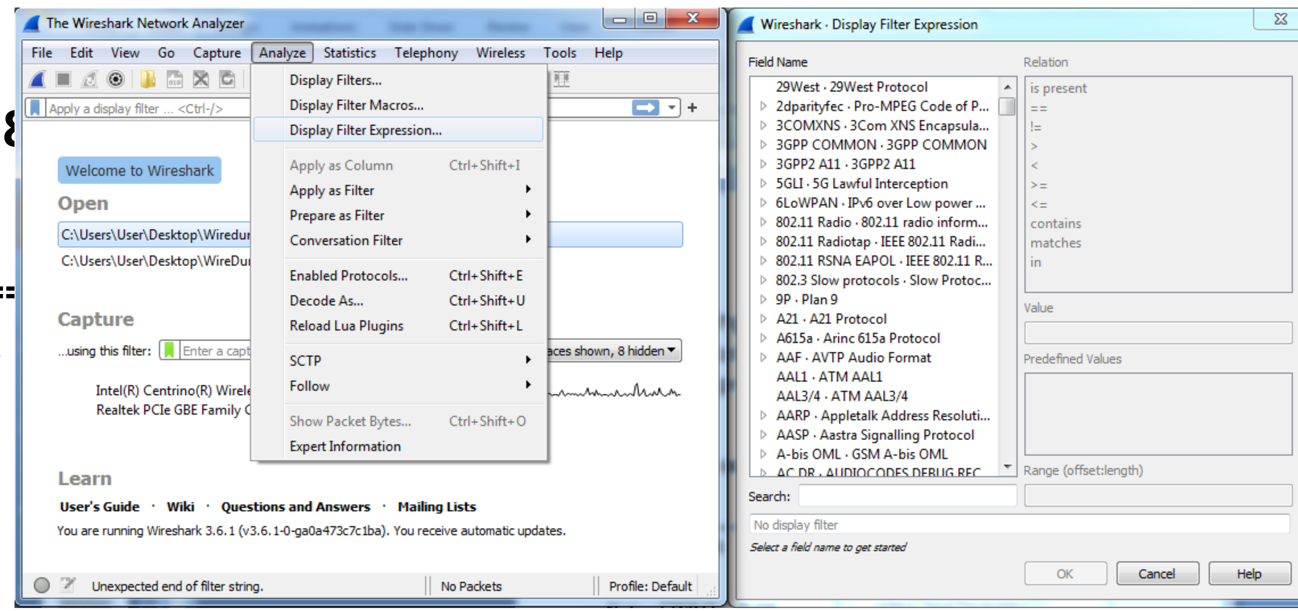



# TCP Segment Structure



# Display Filter

- String1, String2 (Optional settings):
  - Sub protocol categories inside the protocol.
  - Look for a protocol and then click on the "+" character.
  - Example:
  - **tcp.srcport == 8**
  - **tcp.flags == 2**
    - **SYN packet**
    - **Tcp.flags.syn==**
  - **tcp.flags == 18**
    - **SYN/ACK**
  - **TCP Flag field:**




| | | | | | |
|---|---|---|---|---|---|
| U | A | P | R | S | F |
| R | C | S | S | Y | I |
| G | K | H | T | N | N |

# Display Filter Expressions

- snmp || dns || icmp
  - Display the SNMP or DNS or ICMP traffics.
- tcp.port == 25
  - Display packets with TCP source or destination port 25.
- tcp.flags
  - Display packets having a TCP flag.
- tcp.flags == 0x02
  - Display packets with a TCP SYN flag.

Six comparison operators are available:

| English format: | C like format: | Meaning: |
|---|---|---|
| eq | == | Equal |
| ne | != | Not equal |
| gt | > | Greater than |
| lt | < | Less than |
| ge | >= | Greater or equal |
| le | <= | Less or equal |

➡ **Logical expressions**:

| English format: | C like format: | Meaning: |
|---|---|---|
| and | && | Logical AND |
| or | \|\| | Logical OR |
| xor | ^^ | Logical XOR |
| not | ! | Logical NOT |

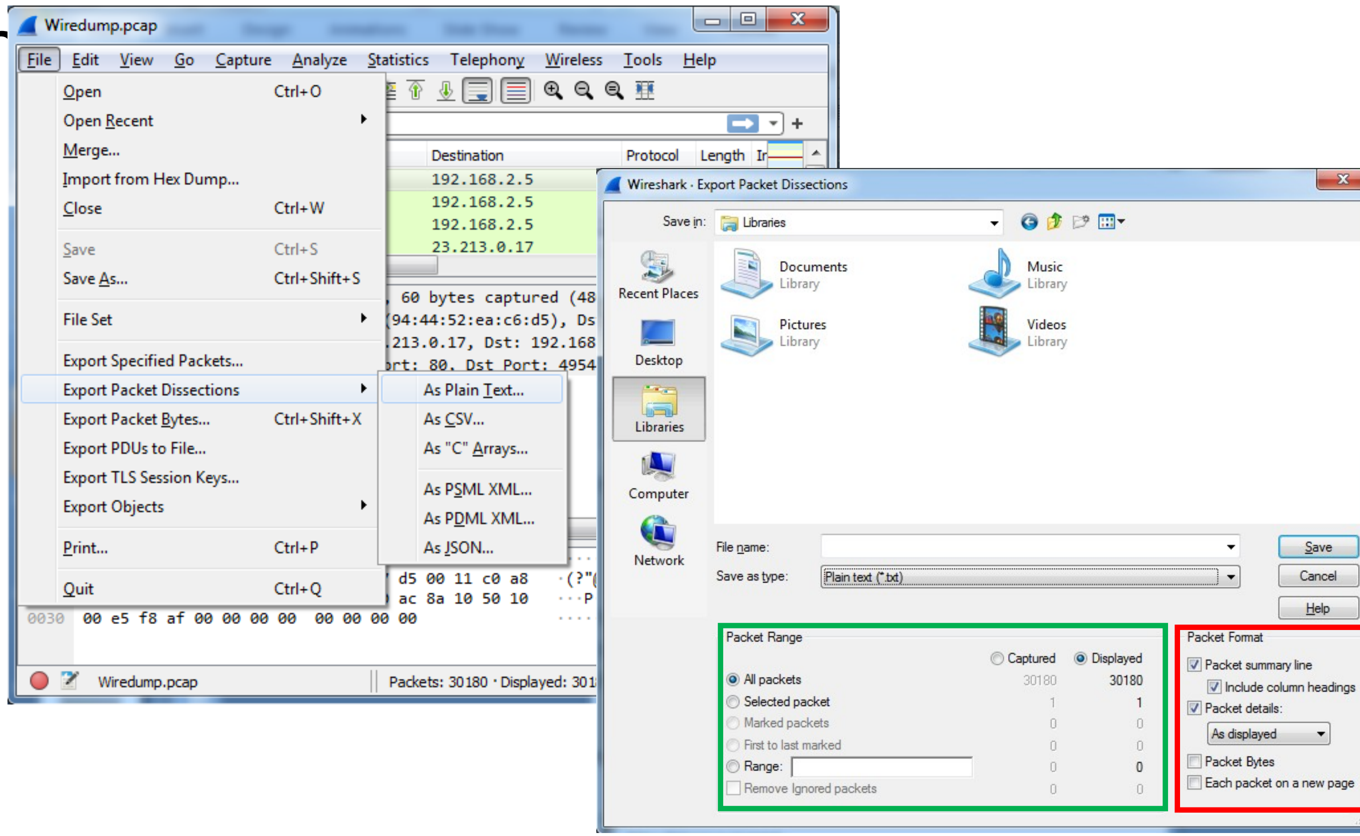
**If the filter syntax is correct, it will be highlighted in green, otherwise if there is a syntax mistake it will be highlighted in red.**

Filter: tcp.port == 100 — **Correct syntax**

Filter: tcp.port = 100 — **Wrong syntax**

# Save Filtered Packets after Using Display Filter

- We can also save all filtered packets in text file for further analysis
- Operation

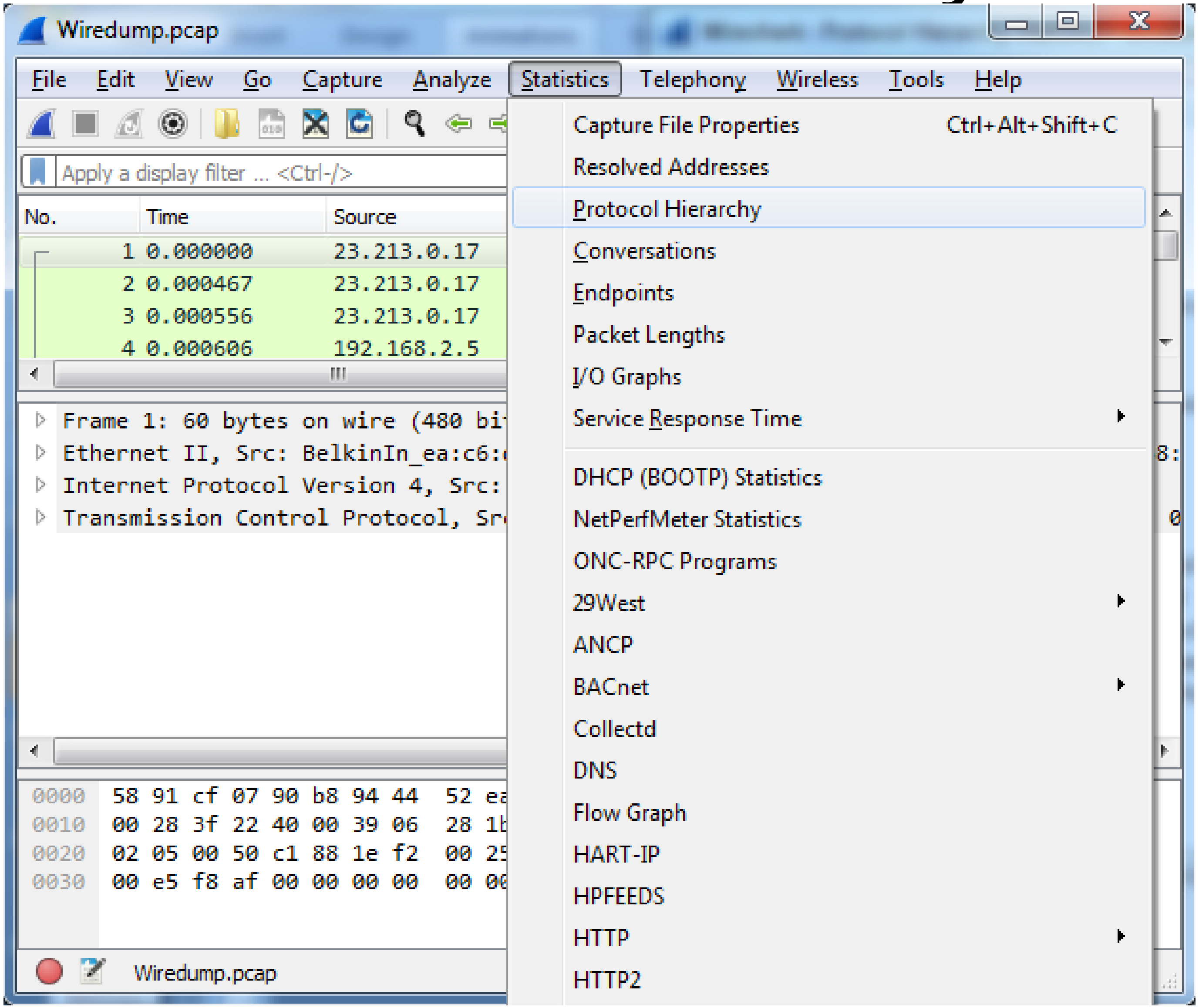



**File→Export packet dissections →as "plain text" file**

**1). In "packet range" option, select "Displayed"**

**2). Choose "summary line" or "detail"**

# Protocol Hierarchy



# Protocol Hierarchy (contd.)

Wireshark · Protocol Hierarchy Statistics · Wiredump.pcap

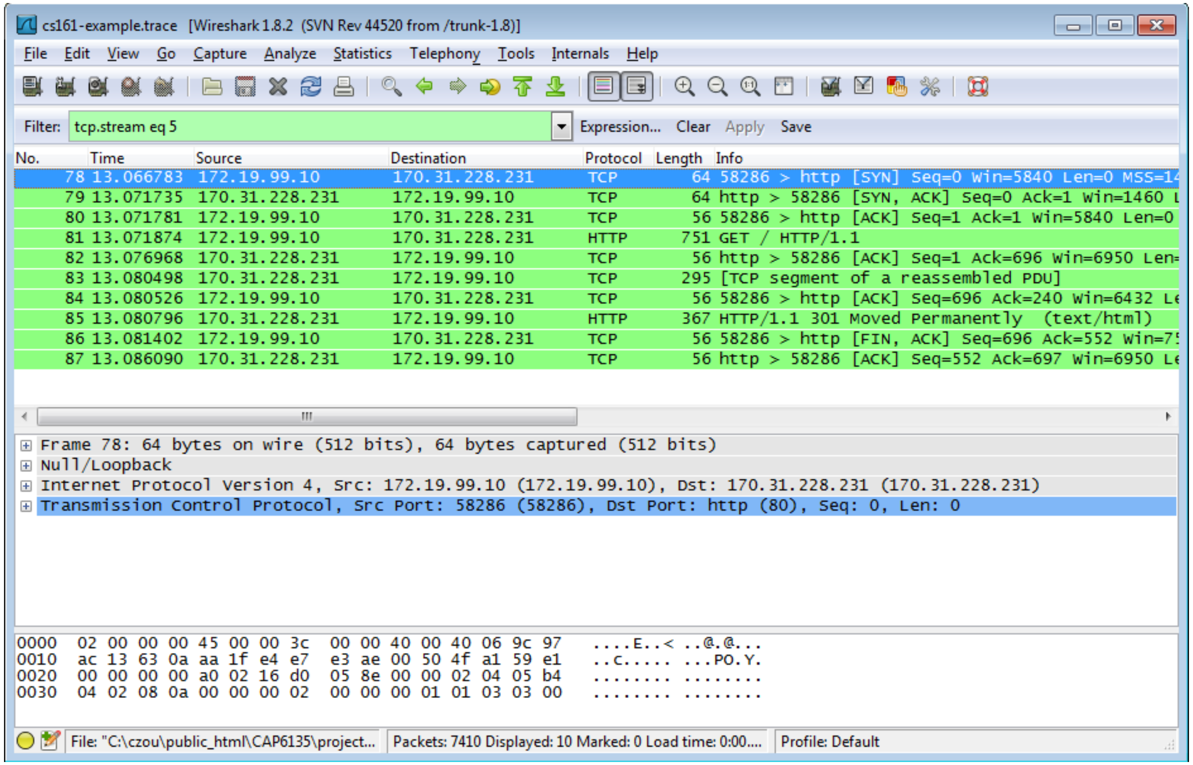
| Protocol | Percent Packets | Packets | Percent Bytes | Bytes | Bits/s | End Packets | End Bytes | End Bits/s |
|---|---|---|---|---|---|---|---|---|
| Frame | 100.0 | 30180 | 100.0 | 26368284 | 5445 k | 0 | 0 | 0 |
| Ethernet | 100.0 | 30180 | 1.6 | 422520 | 87 k | 0 | 0 | 0 |
| Internet Protocol Version 6 | 0.0 | 9 | 0.0 | 360 | 74 | 0 | 0 | 0 |
| User Datagram Protocol | 0.0 | 3 | 0.0 | 24 | 4 | 0 | 0 | 0 |
| DHCPv6 | 0.0 | 3 | 0.0 | 261 | 53 | 3 | 261 | 53 |
| Internet Control Message Protocol v6 | 0.0 | 6 | 0.0 | 192 | 39 | 6 | 192 | 39 |
| Internet Protocol Version 4 | 99.9 | 30160 | 2.3 | 603228 | 124 k | 0 | 0 | 0 |
| User Datagram Protocol | 8.9 | 2691 | 0.1 | 21528 | 4445 | 0 | 0 | 0 |
| Simple Service Discovery Protocol | 0.2 | 63 | 0.1 | 16611 | 3430 | 63 | 16611 | 3430 |
| QUIC IETF | 8.3 | 2496 | 5.8 | 1517524 | 313 k | 2439 | 1471948 | 303 k |
| Malformed Packet | 0.0 | 2 | 0.0 | 0 | 0 | 2 | 0 | 0 |
| NetBIOS Name Service | 0.1 | 23 | 0.0 | 1240 | 256 | 23 | 1240 | 256 |
| NetBIOS Datagram Service | 0.0 | 4 | 0.0 | 765 | 157 | 0 | 0 | 0 |
| SMB (Server Message Block Protocol) | 0.0 | 4 | 0.0 | 437 | 90 | 0 | 0 | 0 |
| SMB MailSlot Protocol | 0.0 | 4 | 0.0 | 100 | 20 | 0 | 0 | 0 |
| Microsoft Windows Browser Protocol | 0.0 | 4 | 0.0 | 93 | 19 | 4 | 93 | 19 |
| Multicast Domain Name System | 0.0 | 4 | 0.0 | 244 | 50 | 4 | 244 | 50 |
| Dynamic Host Configuration Protocol | 0.0 | 3 | 0.0 | 900 | 185 | 3 | 900 | 185 |
| Domain Name System | 0.3 | 95 | 0.1 | 14561 | 3006 | 95 | 14561 | 3006 |
| Data | 0.2 | 58 | 0.0 | 2278 | 470 | 58 | 2278 | 470 |
| Transmission Control Protocol | 91.0 | 27457 | 90.2 | 23775639 | 4909 k | 20838 | 15200882 | 3139 k |
| Transport Layer Security | 22.1 | 6658 | 82.3 | 21701465 | 4481 k | 6513 | 18459368 | 3812 k |

*No display filter.*

Close | Copy | Help

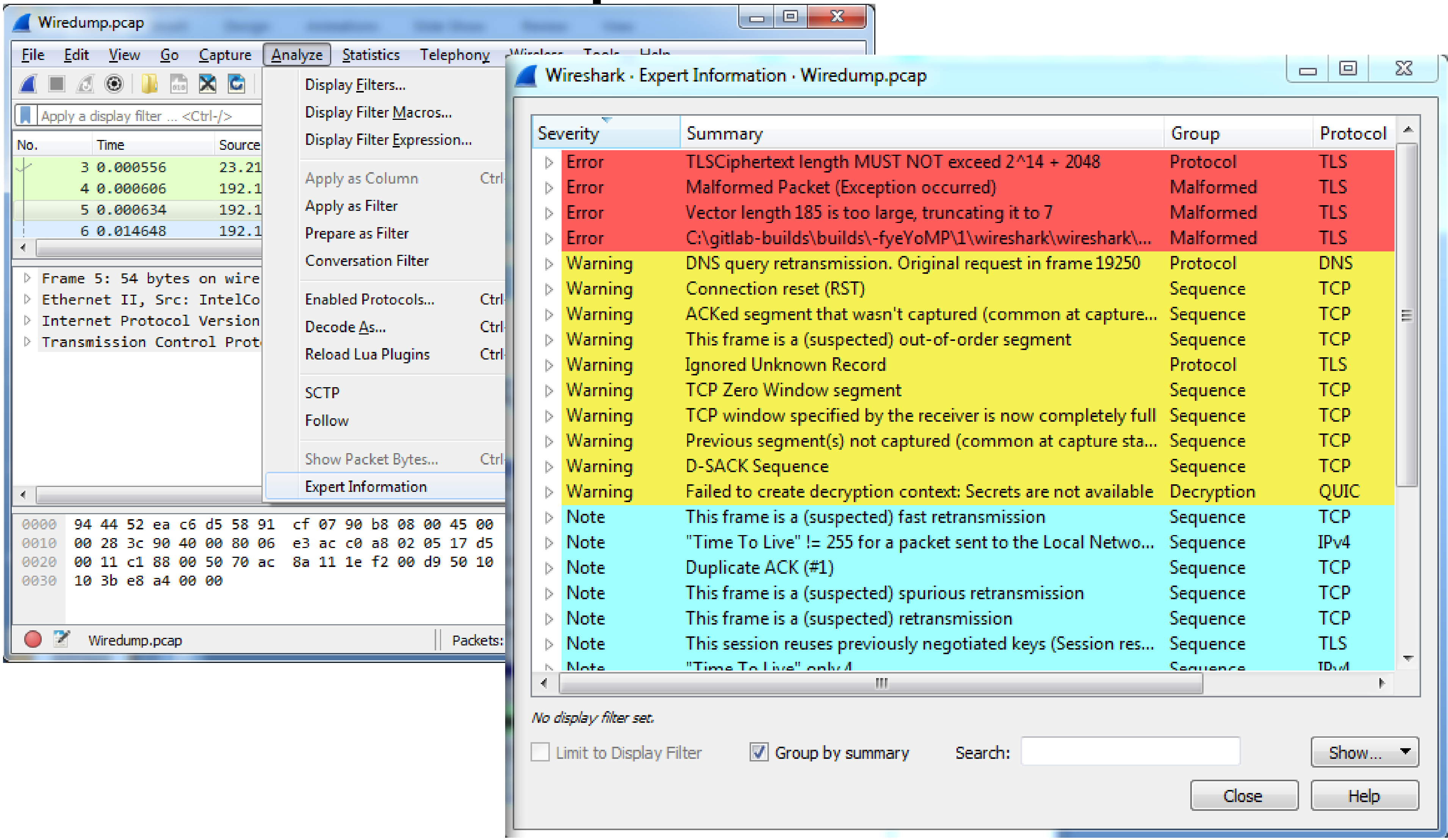
# Filter out/in Single TCP Stream

- When click “filter out this TCP stream” in previous page’s box, new filter string will contain like:
  - http and !(tcp.stream eq 5)
- So, if you use “tcp.stream eq 5” as filter string, you keep this HTTP session

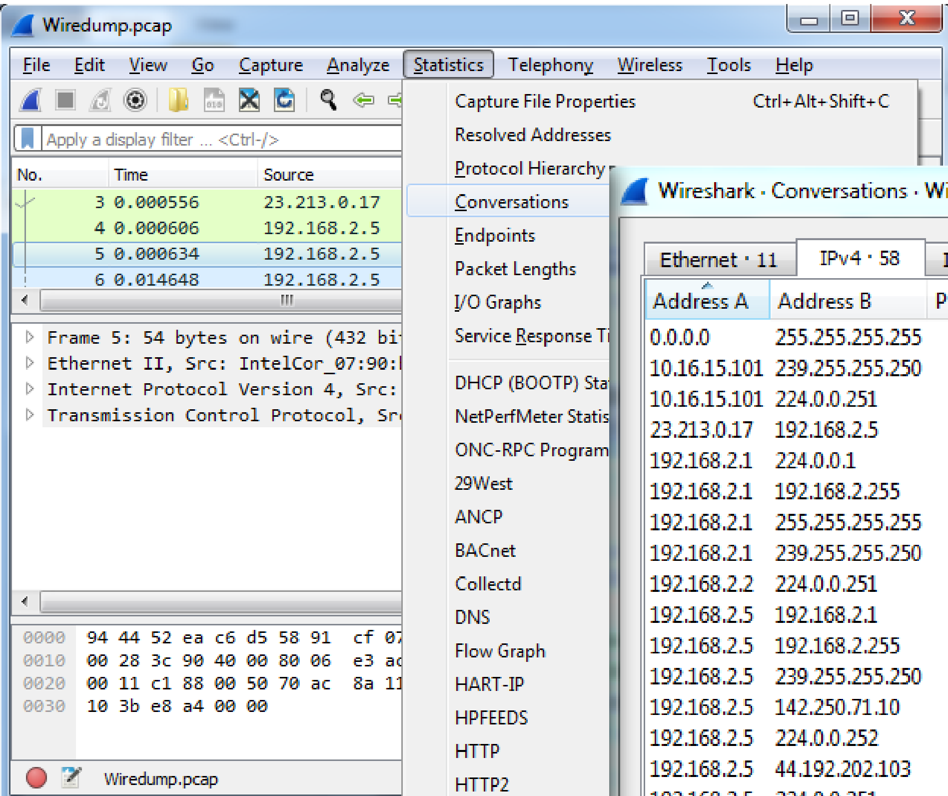


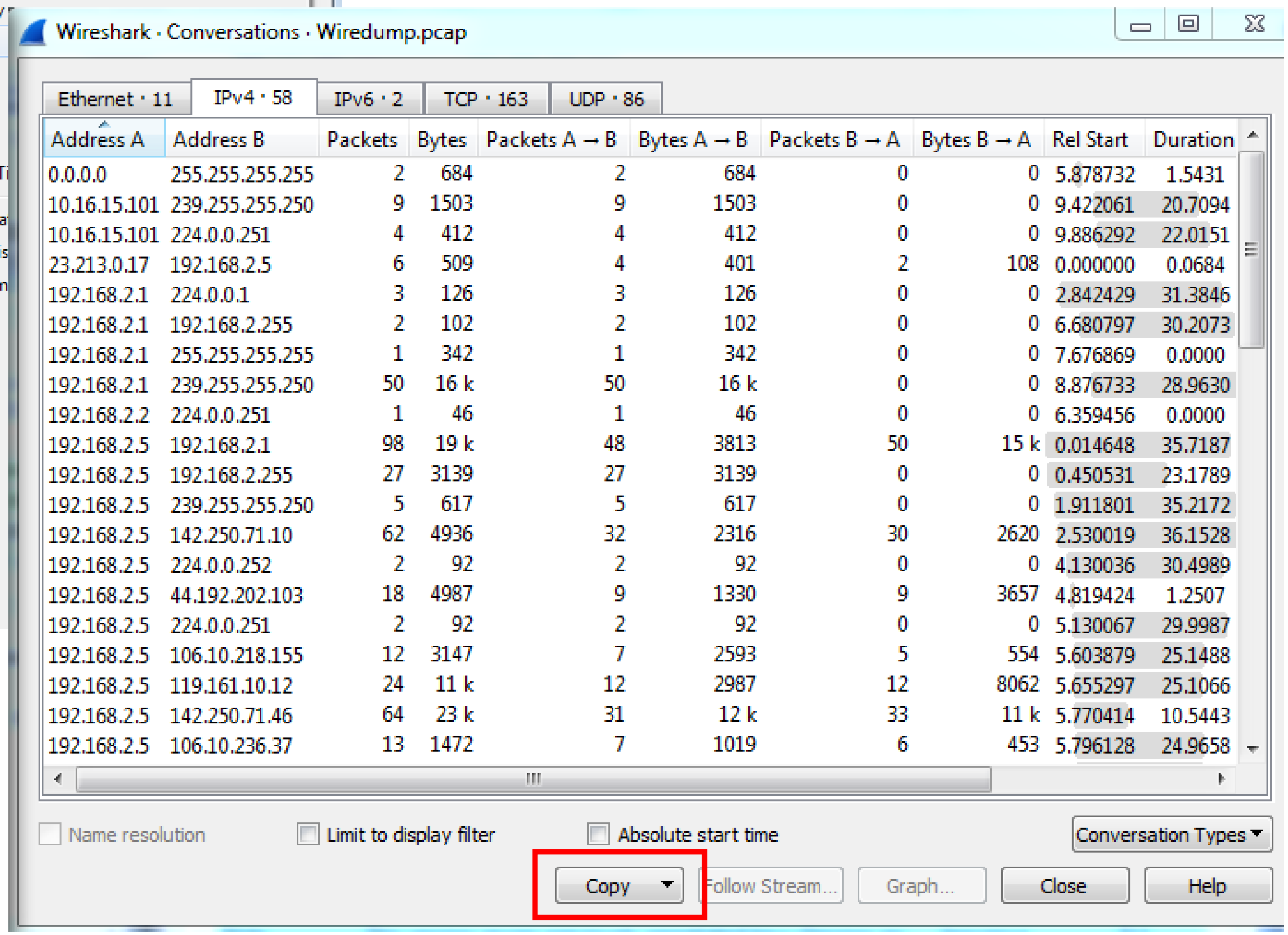

**Old version**

# Expert Info


Wiredump.pcap
File Edit View Go Capture Analyze Statistics Telephony
Apply a display filter ... <Ctrl-/>
No. Time Source
3 0.000556 23.21
4 0.000606 192.1
5 0.000634 192.1
6 0.014648 192.1
Frame 5: 54 bytes on wire
Ethernet II, Src: IntelCo
Internet Protocol Version
Transmission Control Prot
0000 94 44 52 ea c6 d5 58 91 cf 07 90 b8 08 00 45 00
0010 00 28 3c 90 40 00 80 06 e3 ac c0 a8 02 05 17 d5
0020 00 11 c1 88 00 50 70 ac 8a 11 1e f2 00 d9 50 10
0030 10 3b e8 a4 00 00
Wiredump.pcap
Packets:
Display Filters...
Display Filter Macros...
Display Filter Expression...
Apply as Column
Apply as Filter
Prepare as Filter
Conversation Filter
Enabled Protocols...
Decode As...
Reload Lua Plugins
SCTP
Follow
Show Packet Bytes...
Expert Information
Wireshark · Expert Information · Wiredump.pcap
Severity Summary Group Protocol
Error TLSCiphertext length MUST NOT exceed 2^14 + 2048 Protocol TLS
Error Malformed Packet (Exception occurred) Malformed TLS
Error Vector length 185 is too large, truncating it to 7 Malformed TLS
Error C:\gitlab-builds\builds\-fyeYoMP\1\wireshark\wireshark\... Malformed TLS
Warning DNS query retransmission. Original request in frame 19250 Protocol DNS
Warning Connection reset (RST) Sequence TCP
Warning ACKed segment that wasn't captured (common at capture... Sequence TCP
Warning This frame is a (suspected) out-of-order segment Sequence TCP
Warning Ignored Unknown Record Protocol TLS
Warning TCP Zero Window segment Sequence TCP
Warning TCP window specified by the receiver is now completely full Sequence TCP
Warning Previous segment(s) not captured (common at capture sta... Sequence TCP
Warning D-SACK Sequence Sequence TCP
Warning Failed to create decryption context: Secrets are not available Decryption QUIC
Note This frame is a (suspected) fast retransmission Sequence TCP
Note "Time To Live" != 255 for a packet sent to the Local Netwo... Sequence IPv4
Note Duplicate ACK (#1) Sequence TCP
Note This frame is a (suspected) spurious retransmission Sequence TCP
Note This frame is a (suspected) retransmission Sequence TCP
Note This session reuses previously negotiated keys (Session res... Sequence TLS
No display filter set.
Limit to Display Filter
Group by summary
Search:
Show...
Close
Help

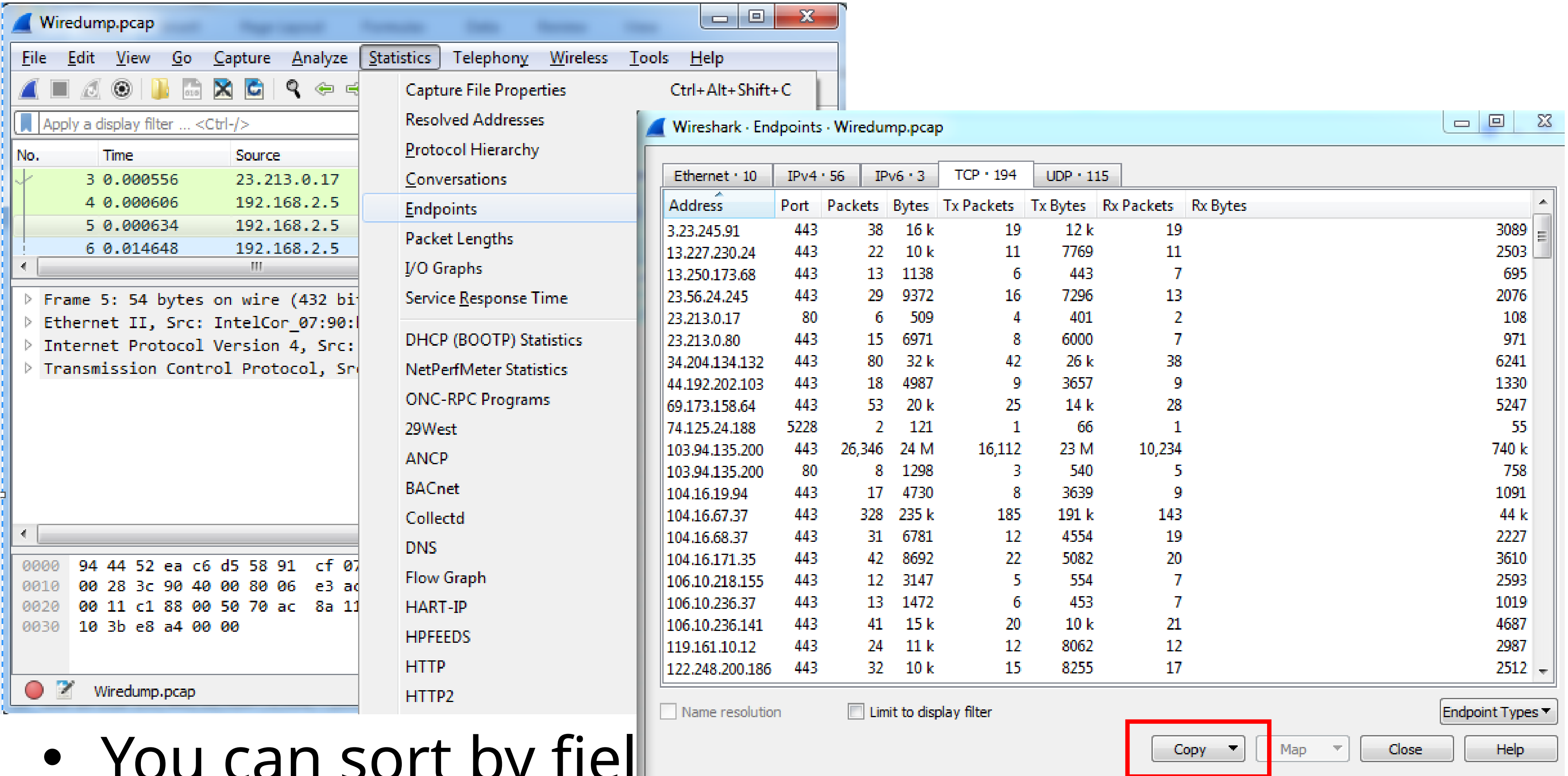
# Conversations




- Use the “Copy” button to copy all text into clipboard (file)
  - Analyze the file to get what statistics you want

# Find EndPoint Statistics




- You can sort by fiel
- Use the "Copy" button to copy all text into clipboard (file)

# On-line Wireshark Trace Files

- Public available .pcap files:
  - http://www.netresec.com/?page=PcapFiles
- http://www.tp.org/jay/nwanalysis/traces/Lab%20Trace%20Files/
- Wiki Sample capture
  - https://wiki.wireshark.org/SampleCaptures

# Example Trace File and Questions

- Network Forensic Puzzle Contests
  - http://forensicscontest.com/2010/02/03/puzzle-4-the-curious-mr-x
- SharkFest'15 Packet Challenge
  - https://sharkfest.wireshark.org/assets/presentations15/packetchallenge.zip

# Source of Installer

- https://www.wireshark.org/download.html

# Thank You

Acknowledgement: Web sources